MOSAIC-Med: Research Proposal

Testing Whether Multiple Small Neural Networks Can Outperform One Large Neural Network for Healthcare Diagnosis Under Fixed Resource Budgets

Research concept: A parameter- and compute-controlled study of model multiplicity, architectural diversity, adaptive expert routing, uncertainty-triggered computation, and robustness in medical diagnosis.

Important novelty principle: The project should not claim that ensembles, mixture-of-experts, diversity regularization, or dynamic inference are individually new. The proposed contribution is their integration into a controlled scientific framework that asks how a fixed neural-network capacity budget should be distributed among one large model versus multiple specialized small models, and whether adaptive collaboration improves clinical robustness and efficiency.

1. Problem Statement

Modern medical AI often follows a scaling strategy: increase the size and capacity of a single neural network to improve predictive performance. However, a separate line of machine-learning research has shown that splitting a fixed parameter budget across multiple smaller models can sometimes outperform a single model with the same total parameter count. Whether this phenomenon transfers reliably to healthcare neural networks—and under which conditions—remains insufficiently characterized.

Healthcare provides an especially demanding setting because predictive performance is not the only requirement. Models must also be calibrated, robust to distribution shift, resilient to noisy or missing measurements, computationally efficient, and capable of identifying cases for which automated prediction is uncertain.

A simple comparison of 'two small networks versus one large network' would not be sufficiently novel because ensemble learning and mixture-of-experts approaches are established. The research problem is therefore reformulated as a controlled investigation of **model multiplicity versus model scale**, followed by a new adaptive framework in which multiple compact experts develop complementary representations, are selectively activated, and can escalate difficult cases to additional experts.

Core research question: Under an equal total parameter and computational budget, can a population of heterogeneous small neural networks provide superior diagnostic performance, calibration, robustness, and resource efficiency compared with a single large neural network?

2. Research Objectives

- Quantify the effect of dividing a fixed neural-network capacity budget among 1, 2, 4, 8, and potentially 16 models.
- Determine whether heterogeneous architectures provide advantages over homogeneous small-model populations.
- Develop an adaptive expert-routing mechanism that selects only the experts relevant to an individual patient.
- Introduce diversity control so that experts learn complementary rather than redundant representations.
- Develop uncertainty-triggered escalation, allowing additional experts to be activated for difficult cases.
- Evaluate robustness to noise, missing inputs, and external/distribution-shift data.
- Compare diagnostic utility against parameter count, FLOPs, latency, memory, and average number of active experts.

3. Proposed Solution: MOSAIC-Med

MOSAIC-Med (working name: Model Splitting with Adaptive, Specialized, Inhibitory and Collaborative Experts for Medical AI) is a population-based neural architecture designed to investigate whether model diversity can substitute for individual model scale.

The system contains multiple compact experts. Rather than forcing every expert to process every case and simply averaging their outputs, a learned router estimates which experts are useful for a particular patient. A diversity mechanism discourages unnecessary representational duplication, while an uncertainty mechanism can activate additional experts when the initial prediction is unreliable.

Component	Purpose
Expert population	Multiple compact neural networks whose combined parameter count is controlled.
Heterogeneous inductive biases	Use complementary architectures/representations rather than identical copies.
Adaptive router	Selects or weights experts on a patient-specific basis.
Diversity regularization	Reduces harmful redundancy while preserving predictive performance.
Failure-aware specialization	Encourages complementary strengths based on observed error patterns.
Uncertainty-triggered escalation	Activates additional experts only when confidence/calibration criteria indicate that more computation is justified.
Abstention pathway	Allows the system to flag persistently uncertain cases for human review instead of forcing a diagnosis.

4. Proposed Experimental Methodology

4.1 Clinical testbed

The first implementation is proposed for 12-lead ECG diagnosis using a public dataset such as PTB-XL, with an independent external ECG dataset reserved for external validation. The same framework can later be transferred to medical imaging or multimodal clinical data. The dataset choice should be finalized after a current literature and data-quality review.

4.2 Fixed-budget experimental design

Condition	Number of models	Approx. size/model	Total parameter budget
Large baseline	1	32M	32M
Split-2	2	16M	32M
Split-4	4	8M	32M
Split-8	8	4M	32M
Split-16	16	2M	32M

The exact budget should be selected after profiling the dataset and hardware. In addition to parameter matching, a second analysis should match or report inference FLOPs/latency so that a parameter advantage is not confused with a compute advantage.

4.3 Expert diversity

- Homogeneous condition: all experts use the same architecture.
- Heterogeneous condition: experts use complementary architectures such as 1D CNN/ResNet, TCN, lightweight Transformer, and frequency-domain or multi-scale temporal branches.
- Representation diversity should be measured rather than assumed.

4.4 Training objective

A conceptual objective is: $\mathbf{L = L_{diag} + \lambda_d L_{div} + \lambda_r L_{route} + \lambda_c L_{cal} + \lambda_f L_{fail}}$.

Ldiag optimizes diagnostic prediction. **Ldiv** discourages harmful representation redundancy. **Lroute** encourages sparse and meaningful expert selection. **Lcal** supports reliable probabilities. **Lfail** encourages complementary error patterns. These terms should be introduced incrementally in ablation experiments rather than assuming that all terms improve performance.

4.5 Patient-adaptive inference

Stage 1: activate a small number of experts. Stage 2: estimate predictive uncertainty and expert disagreement. Stage 3: if the case is sufficiently uncertain, activate additional experts. Stage 4: if uncertainty remains above a prespecified threshold, abstain and flag the case for human review.

This creates an adaptive computation mechanism: easy cases consume less computation, while difficult or distribution-shifted cases receive additional model capacity.

5. Validation Framework

Domain	Measures	Purpose
Discrimination	AUROC, AUPRC, sensitivity, specificity, F1, balanced accuracy	Diagnostic performance, including class-imbalance-sensitive evaluation.
Calibration	Brier score, ECE, reliability diagrams	Whether predicted probabilities correspond to observed risk.
Uncertainty	Selective risk, coverage, abstention rate	Whether uncertainty-based escalation/abstention is useful.
Robustness	Noise, baseline drift, missing leads/features, perturbations	Sensitivity to realistic data degradation.
External validity	Independent external dataset	Generalization across acquisition/population/domain differences.
Efficiency	Parameters, FLOPs, latency, peak memory, active experts	Resource cost of the competing strategies.
Fairness/subgroups	Performance and calibration by relevant documented subgroups	Identify whether gains are consistent across populations.

6. Statistical Analysis Plan

- **Primary comparison:** compare paired patient-level predictions between the large model and the proposed multi-expert system using paired methods appropriate to each metric.
- **AUROC:** DeLong's test for correlated ROC curves when comparing predictions on the same test subjects.
- **Binary paired outcomes:** McNemar's test for paired classification disagreements at a prespecified operating threshold.
- **Bootstrap confidence intervals:** stratified patient-level bootstrap, preferably 2,000–10,000 resamples, for AUROC, AUPRC, sensitivity, specificity, F1, Brier score, and differences between systems.
- **Multiple configurations:** because 1/2/4/8/16 experts create repeated comparisons, control multiplicity using Holm correction or a prespecified hierarchical testing procedure.
- **Multiple random seeds:** train each configuration across at least 5 independent seeds. Report mean ± SD and confidence intervals across runs, while keeping patient-level test predictions separate from seed-level variability.
- **Repeated-measures analysis:** if many configurations and repeated seeds are compared, use a mixed-effects model with configuration as a fixed effect and seed/dataset split as appropriate random effects; alternatively use a preregistered non-parametric approach.
- **Efficiency:** compare latency/FLOPs using paired measurements where possible; report effect sizes and confidence intervals rather than only p-values.
- **Effect size:** report absolute AUROC/AUPRC differences, calibration differences, relative compute reduction, and standardized effect sizes where appropriate.
- **Significance:** use a prespecified α (e.g., 0.05), but interpret effect sizes and confidence intervals alongside p-values. Statistical significance alone should not determine clinical relevance.

7. Critical Ablation Studies

- 1 vs 2 vs 4 vs 8 vs 16 experts at fixed total parameters.
- Homogeneous vs heterogeneous experts.
- Simple averaging vs learned fusion vs sparse adaptive routing.
- Without vs with diversity regularization.
- Without vs with failure-aware specialization.

- Without vs with uncertainty-triggered escalation.
- Without vs with abstention.
- Random routing vs learned routing.
- Parameter-matched vs FLOP-matched comparisons.
- In-domain test performance vs external-domain performance.

8. What Would Count as Evidence for the Hypothesis?

Evidence supporting the hypothesis would be a reproducible improvement of the multi-expert approach over the parameter-matched single large network across clinically relevant metrics, with confidence intervals excluding negligible effects, and with the advantage remaining after controlling for random seed, compute, and multiple comparisons.

A negative result is also scientifically valuable. If increasing the number of small experts eventually hurts performance, the experiments can identify the transition point at which individual expert capacity becomes insufficient. This prevents the study from being designed merely to confirm the motivating claim.

9. Reproducibility and Leakage Controls

- Split at the patient level, never at the individual ECG/recording level when multiple records from the same patient exist.
- Keep the external dataset completely untouched during model development.
- Prespecify primary endpoint, primary comparison, statistical tests, and exclusion criteria before final evaluation.
- Use identical preprocessing and data augmentation rules across competing model families unless an ablation explicitly tests them.
- Release code, configurations, random seeds, model checkpoints where licensing permits, and exact preprocessing steps.

10. Expected Scientific Contribution

The project is designed to answer a broader question than whether an ensemble is accurate: **how should a fixed neural-network capacity budget be distributed between model scale, model multiplicity, diversity, and adaptive computation for medical diagnosis?**

A successful outcome would provide an empirical scaling curve across the number of experts, characterize when multiplicity helps or hurts, determine whether architectural diversity contributes beyond model count, and test whether adaptive activation can convert a many-model system into an efficient patient-specific inference mechanism.

11. Proposed Paper Structure

- Introduction — model scaling versus model multiplicity in healthcare.
- Related work — ensembles, mixture-of-experts, model splitting, diversity, uncertainty, and medical AI robustness.
- Research questions and hypotheses.
- MOSAIC-Med architecture and mathematical formulation.
- Fixed-budget experimental protocol.
- Results: scaling with number of experts.
- Ablations: diversity, routing, inhibition, escalation, and abstention.
- Robustness and external validation.
- Statistical analysis and efficiency trade-offs.
- Discussion of when small-model multiplicity can or cannot replace individual model scale.

Note on novelty: This document is a research-design proposal, not a guarantee that every individual mechanism is unprecedented. Before claiming novelty in a manuscript, conduct a systematic literature search across PubMed, IEEE Xplore, Scopus/Web of Science, Google Scholar, and relevant ML proceedings, using combinations of terms such as fixed-budget ensembles, model splitting, small-vs-large neural networks, medical mixture-of-experts, diversity regularization, adaptive routing, uncertainty-triggered inference, and external medical-domain shift.